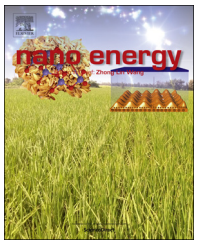
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COMMUNICATION

High-output current density of the triboelectric nanogenerator made from recycling rice husks

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Abstract
This work, we are the first to discover the high-output current density of the triboelectric nanogenerator (TENG) using rice husks as a source material. The raw rice husks (RH) can be directly phase transited into the amorphous SiO_2 (RH_{SiO2}) structure with highly nanoporous fragments by the thermal annealing with additional acid hydrolysis process. The RH_{SiO2}-TENG's configuration is designed by polytetrafluoroethene (PTFE) and RH_{SiO2} films, which are chemically and thermally more stable than the metallic film. The pore size around 20–40 nm is widely distributed throughout the SiO_2 fragments that possess rich Si-O-Si and OH stretching bonds with strong tendency of repulsing electron because the H atoms have an extremely low electron affinity, leading to the RH_{SiO2} film exhibits much lower electron affinity when compared with the commercial SiO_2 nanoparticles. As a consequence, the area power density of the RH_{SiO2} triboelectric nanogenerator (RH_{SiO2}-TENG) reaches 0.84 W m^{-2} with a peak short-circuit current density of 5.7 mA m^{-2}. Such a short-circuit current density is almost among the highest reported value based on dielectric-to-dielectric mode triboelectric nanogenerators. Rice husk possess many advantageous traits such as their light weight, low cost, being environmentally friendly, high porosity, excellent robustness, exceptionally chemical and thermal stability for superior corrosion resistance, makes it a valuable material for industrial applications.
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Introduction

The rapidly expanding personal electronics in communication, wearable electronics, medical devices, environmental monitoring have dramatically increasing in the past decade. The number of batteries in the waste stream increase with increasing the number of portable electronics used [1]. It is very critical task to search a better solution to reduce the electronic waste especially on the dispose of batteries. Researchers are now working to find alternative energy such as solar cell [2], biofuels [3], thermoelectricity [4,5], and nanogenerators [6] to provide the electricity for personal electronics [7]. Among them, the nanogenerator first discovered by Wang's group, is offering a promising solution for solving the energy of portable devices with reducing concerns of dispose batteries and other external power sources [8]. The key components inside a nanogenerator can be divided into three kind of source materials such as piezoelectric [9,10], triboelectric [11], and pyroelectric materials [12,13], which have been demonstrating in a variety of applications for detecting environmental change and electronic devices such as wireless sensing [12], self-powered active sensors [14-16], force sensors [17-19], and human-machine interfacing devices [20-22]. Lately, triboelectric nanogenerators (TENGs) have been attracting much attention for researchers due to their high energy conversion efficiency and low production cost. Most TENGs' materials are organic matters, which can readily be expanded to a large-scale fabrication for prospective industrial applications [23]. The working principle of TENG usually requires two surfaces with opposite charges between each other. By utilizing the triboelectric effect and electrostatic induction, the TENG can efficiently convert the irregularly mechanical energy into electricity. Thus, establishing a large triboelectric potential difference between two charge surfaces is very important task in order to obtain a high output power from TENG.

There have mainly two attached-electrode contact modes in TENGs such as dielectric-to-conductor and dielectric-to-dielectric modes [24]. Most TENGs' configurations were designed by PTFE and metallic film (ie., aluminum film) because the charge transfer efficiency from dielectric-to-conductor mode is higher than that from dielectric-to-dielectric mode, indicated that dielectric-to-conductor TENG is more advantageous for high-output power nanogenerators [18].

In the triboelectric series table, metallic film is positively charged ions to absorb negative ions [18] while polytetrafluoroethen (PTFE) serves as the most negatively charged ions because it is terminated with the most electronegative functional group-fluoro (F) [25]. To generate a remarkable difference of triboelectric potential between two surfaces in dielectric-to-conductor TENG, tremendous efforts have been dedicating to create a high surface to volume ratio on the surface of the PTFE and metallic film through a variety of nano-technology methodologies. However, considering the long-time stability issue especially for the energy harvesting from fluid flow in water (ie., ocean waves), the dielectric-to-dielectric TENG is offering thermally and chemically more stable than the dielectric-to-conductor TENG because corrosion of metal in the presence of water is a common problem. Moreover, the short-circuit current density still plays an important role in order to process the TENG for commercial applications. Thus, developing a highly positive and negative charge of triboelectric surfaces in enhancing the smooth separation and increasing the friction area between two surfaces become very critical tasks [26].

This work, we are the first to discover a high short-circuit current density of triboelectric nanogenerator using recycling rice husks as a source material. After leached the rice husks in hydrochloric acid (HCl) solution and calcined them at 700 °C for 3 h, the purely nanoporous silica powder with pore sizes around 20-40 nm could be produced [24] as denoted by RH_{SiO2}. The structure of the pores and pore size distribution are greatly dominated by the nature of the raw rice husks. Such nanoporous structures with greatly positively charged surfaces are highly distributed throughout the rice-husk SiO_2 (RH_{SiO2}) fragments. Thus, the nanoporous RH_{SiO2} and untreated polytetrafluoroethylene (PTFE) films are served as two dielectric layers of TENG at the opposite ends of the triboelectric series. On the basis of our FTIR spectra, the nanoporous RH_{SiO2} fragments offers highly dense Si-O-Si, Si-OH, and OH stretching bonds compare to commercial SiO_2 nanoparticles. The RH_{SiO2} film therefore exhibits strongly tendency to repel the electron because the H atoms have an extremely low electron affinity compared to the PTFE film. The highly positively charged surfaces of the nanoporous RH_{SiO2} film is five times higher than that of commercially non-porous SiO_2 film based on our theoretical calculated results.

Interestingly, the porous RH_{SiO2} film can offer a rapidly charging and discharging process due to its wide range of pore size distribution, leading to the peak short-circuit (I_{sc}) current density of 5.7 mA m^{-2} was measured from the RH_{SiO2} triboelectric nanogenerator (RH_{SiO2} TENG). Such high current density is almost among the highest reported value based on dielectric-to-dielectric mode TENGs. The area power density reaches 0.84 W m^{-2}. The RH_{SiO2} TENG can be used as a direct power source for driving over than 150 of serially connected light-emitting diodes (LED) bulbs. The rice husk is one of the highest-volume agricultural residues, which is 10^8 t of RHs per year across the globe [27]. The dielectric-to-dielectric contact mode of RH_{SiO2} TENG has exceptionally thermal and chemical stability for excellent corrosion resistance, which can be integrated in various substrates and makes it a valuable material for a durably large-scale TENG.

Experimental method

Material preparation

The raw rice husks (RHs) were washed by deionized (DI) water and dried by oven for 60 min to clean their surface's contaminations. To prepare the nanoporous rice-husk SiO_2 (RH_{SiO2}) nanoparticles, the clean rice husks were leached in HCl solution for 2.5 h to remove the metal impurities such as K^+, Na^+, and Ca^{2+} in the raw RHs. Next, the leached RHs were washed with deionized water and dried at 100 °C for one hr and calcined them at 700 °C for 3 h to produce the highly nanoporous distributed in SiO_2 fragments.

Fabrication of the RH-TENGs, ARH-TENGs, and RH_{SiO2}_TENGs

To fabricate the RH-TENGs, the PTFE film was served as a negatively charged electrode. The copper (Cu) film was deposited on one side of the PTFE film to act as a conducting electrode. To prepare another positively charged electrode, first polydimethylsiloxane (PDMS) was spin-coated onto a Cu-coated PET film. After cured the PDMS film at 80 °C for 30 min, the raw rice husk was carefully transferred onto the PDMS film's surface and dried at 100 °C for 30 min, causing the single rice husk was tightly bonded on the surface of the PDMS film. Subsequently, the single rice husk/PDMS/Cu-coated PET substrate was fixed on a glass substrate to act as a positively charged electrode of the SRH-TENG. The effective area of single-rice husk TENG is 2×3 mm^2. Similarly, total 84 grains of rice husks were carefully manipulated to attach on the surface of the PDMS to fabricate the rice-husk arrays TENG (ARH-TENG). In addition, to fabricate the $RH_{SiO2}-$TENGs, the rice husks were leached in hydrochloric acid (HCl) solution. After that, the leached rice husks were thermally annealing at the temperature of 700 °C for 3 h to dissociate the organic matters and produced the pure SiO_2 (named as RH_{SiO2}) fragments with naturally meso- and/or nano-porous structures, as shown the schematic diagram in Figure 1. The highly porous fragments were then milled into the nanoporous RH_{SiO2} particles and transferred them onto the surface of PDMS film to fabricate the $RH_{SiO2}-$TENG.

Material characterization and electrical measurement

The field emission scanning electron microscope (FESEM, HITACHI S4800) was employed to investigate the surface's morphologies of the rice husks. The chemical compositions of the raw rice husks and the RH_{SiO2} films were investigated by high-resolution x-ray photoelectron spectroscopy (ULVAC-PHI 5000). The chemical bonding of the RH_{SiO2} film was conducted by Fourier-transform infrared spectrometer (FTIR, Bruker, Vertex 80 v). The output open-circuit voltage and short-circuit current of the TENGs were measured using a Keithley 6514 programmable electrometer and a low-noise current preamplifier (Stanford Research System Model SR570).

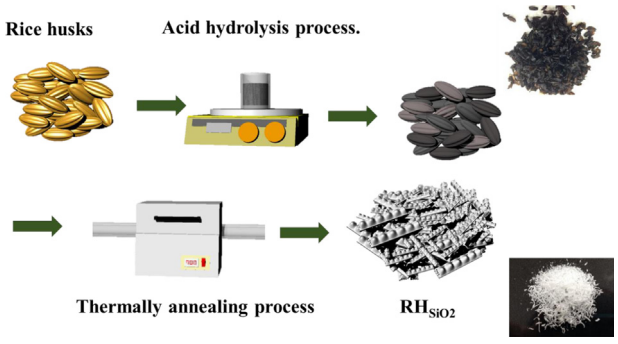


Figure 1 The schematic diagram of the fabrication process for RH_{SiO2} fragments.

Results and discussions

The photograph of raw rice husks is shown in the inset image of Figure 2a. Figure 2a shows that the raw rice husks exhibit uniquely bump-like arrays. The size of bump is around 20 μm. The cross-sectional image of the rice husk is shown in Figure 2b, revealing that the rice husk possesses naturally porous structure with the pore size distribution in the range of 5–10 μm. The inset image of Figure 2b shows that the chemical composition (energy dispersive X-ray spectroscopy, EDS) of the raw rice husk consists of Si and O with atomic ratio of 63 and 37 at%. After the raw rice husks were leached in hydrochloric acid (HCl) solution and thermally decomposed at 700 °C for 3 h. Figure 2c shows that the bump-like arrays of the rice husks did not remarkably change, but the bump's size becomes smaller and broken into numberless pieces. Its cross-sectional image in Figure 2d reveals that a wide range of pore size distribution in the RH_{SiO2} fragments ranging from microscale to nanoscale. The mesoporous size distribution is in the range of 2–5 μm while the nanoporous size distribution is around 2–40 nm (see Figure 2d and its inset upper-right corner). Such porous structures are highly distributed throughout the rice husks. On the basis of the results by EDS (inset Figure 2d lower-left corner) and X-ray photoelectron spectra (see Supplementary material of Figure S1), the composition of rice husk consist of Si and O with an atomic ratio of ~1:2, suggesting that the pure SiO_2 fragments (RH_{SiO2}) with the naturally nanoporous configuration have been formed through the thermally annealing and acid hydrolysis treatment process.

To comprehensively understand the difference of the triboelectric potential in a variety of pore numbers in the RH_{SiO2} films, the theoretical calculations were carried out using the COMSOL software package according to the practical rice husk's morphology with an area of 5×5 cm^2 [28]. The inset Figure 3a shows the schematic diagram of flat SiO_2 and untreated PTFE films, the corresponding triboelectric potential can reach ~253 V. However, Figure 3b shows that the triboelectric potential of the RH_{SiO2} film is increased to 634 V as creating bump arrays on the surface of the SiO_2 film (see Figure 3b inset). When the porous structures were formed in the bump-arrays SiO_2 film (see Figure 3c inset), the corresponding triboelectric potential was significantly increased to 1616 V, as shown in Figure 3c. Figure 3d further shows that the triboelectric potential reaches 2336 V with increasing the number of porosity in the bump-arrays SiO_2 film (see Figure 3d inset). It should be noted that the positively charged density in the raw rice-husk surface is much lower than that in the nanoporous RH_{SiO2} film based on our theoretically calculated results. This is ascribed to that the RH_{SiO2} film raised the dielectric constant from ~2.7 of the raw rice husks to ~3.9. Higher dielectric constant suggests that the RH_{SiO2} film may provide more induced electrostatic charges as resulted from the triboelectric effect [29].

Figure 4a shows the schematic diagram of the individually raw rice-husk TENG. The open-circuit voltage and short-circuit current is around 9 V and 0.05 μA, as shown in Figure 4b and c, respectively. The voltage and current outputs are able to power single LED bulb (see Figure 4d). By integrating two and three RH-TENGs in parallel connections, which can produce the V_{oc} of 14.3 and 19 V, respectively (see Figure 4e). Similarly, the corresponding short-circuit current for these RH-TENGs can be

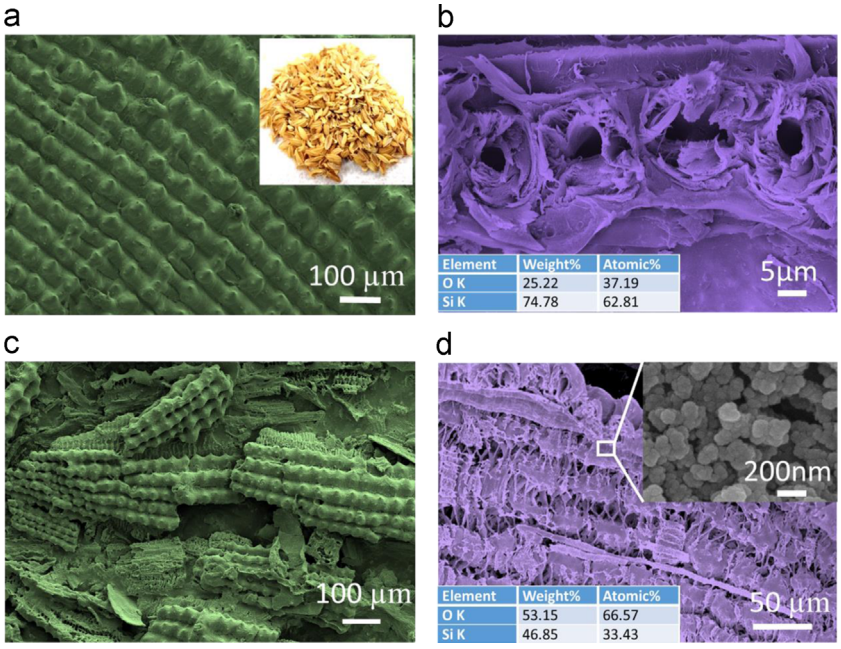


Figure 2 (a) Topographic FESEM image of the RH showing the bump-like arrays. Inset is the photograph of the raw rice husks. (b) Cross-sectional view of rice husk. Inset table is the corresponding chemical composition ratio of rice husk, as analyzed by EDS. (c) The topographic image of the RH_{SiO2} after thermally annealing the raw rice husks. (d) The cross-sectional view of RH_{SiO2} fragments showing the mesoporous structure. Inset table is the corresponding chemical composition ratio. Inset image of the upper-right corner revealing the nanoporous structure in the RH_{SiO2}..

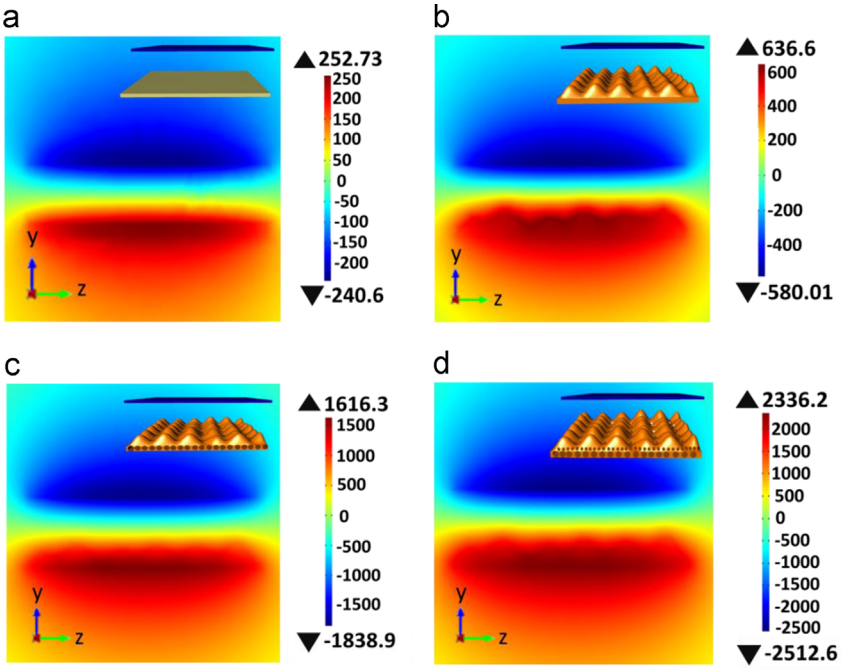


Figure 3 Theoretically calculated results of the output triboelectric potential distribution in the PTFE and RH_{SiO2} films. The simulation was based on the practically rice-husk's surface and the dielectric constant of SiO_2. (a) PTFE film and flat SiO_2 surface (b) PTFE and non-porous RH_{SiO2} films. (c) PTFE and nanoporous RH_{SiO2} films. (d) PTFE and highly nanoporous RH_{SiO2} films.

superimposed (see Figure 4f). The short-circuit current and open-circuit voltage of the individually rice-husk TENG under the different loading resistance is shown in Figure 4g. The instantaneous current decreases with increasing loading resistance due to Ohmic loss while the voltage increases. Consequently, Figure 4h shows that the instantaneous power output of the SRH-TENG reaches a maximal value of ~125 nW under a loading resistance of $1.6\times10^8\Omega$. The total 84 grains of rice

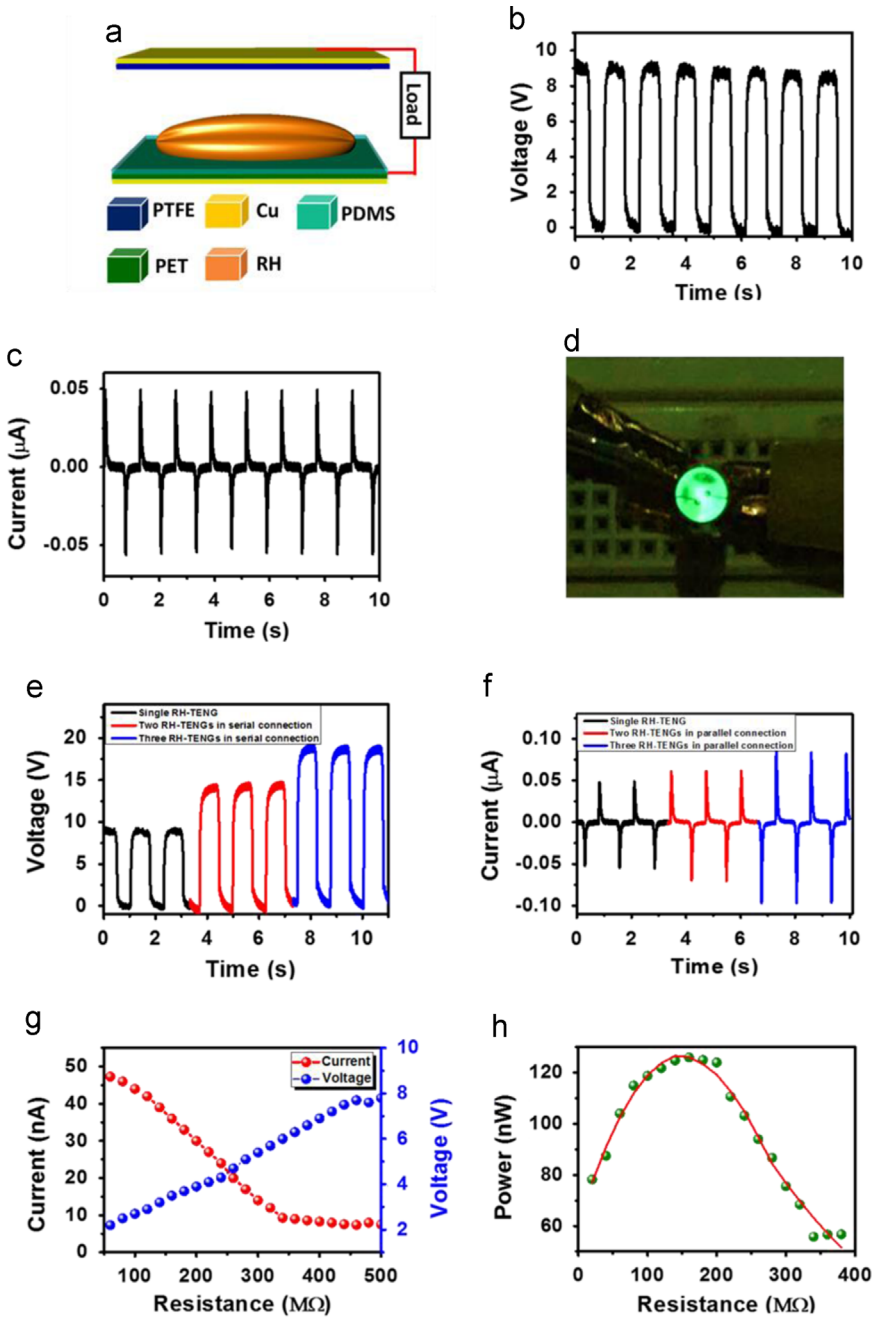


Figure 4 (a) The schematic diagram of SRH-TENG. (b) Open-circuit voltage of the SRH-TENG. (c) Short-circuit current of the SRH-TENG. (d) The photograph for lighting a LED by the SRH-TENG. (e) Open-circuit voltage for three SRH-TENGs in parallel connections. (f) Short-circuit current of the SRH-TENGs in parallel connection. (g) Dependence of the output current and voltage of the SRH-TENG on the external loading resistance. (h) Output power as a function of variety resistance.

husks were carefully transferred to implant on the surface of the PDMS film to fabricate the ARH-TENG, as shown the schematic diagram in Figure 5a. The short-circuit current and open-circuit voltage of the ARH-TENG are dramatically increased up to 1 μA and 190 V, as shown in Figure 5b and c, respectively. Figure 5d reveals that with increasing the loading resistance, the maximal current decreases because of Ohmic loss. In contrast, the maximal output voltage across the load exhibits an opposite trend. Therefore, Figure 5e presents that the maximal output power of ARH-TENG is significantly increased to ~59 μW, which is able to drive over than 50's commercial LED bulbs (see Figure 5f) and the corresponding area power density reaches ~0.12 W m^{-2}.

The raw rice husks can be phase transited into the highly nanoporous SiO_2 fragments with high surface to volume ratio because their related compounds such as celluloses, hemicelluloses, and lignin in raw rice husks were dissociated by the thermally annealing with additional acid hydrolysis processes. Fig. 6a is the schematic diagram of the highly porous RH_{SiO2} TENG. Figure 6b and c shows that a short-circuit current and an open-circuit voltage of the RH_{SiO2} TENG reaches 14 μA and 270 V, respectively. Figure 6d

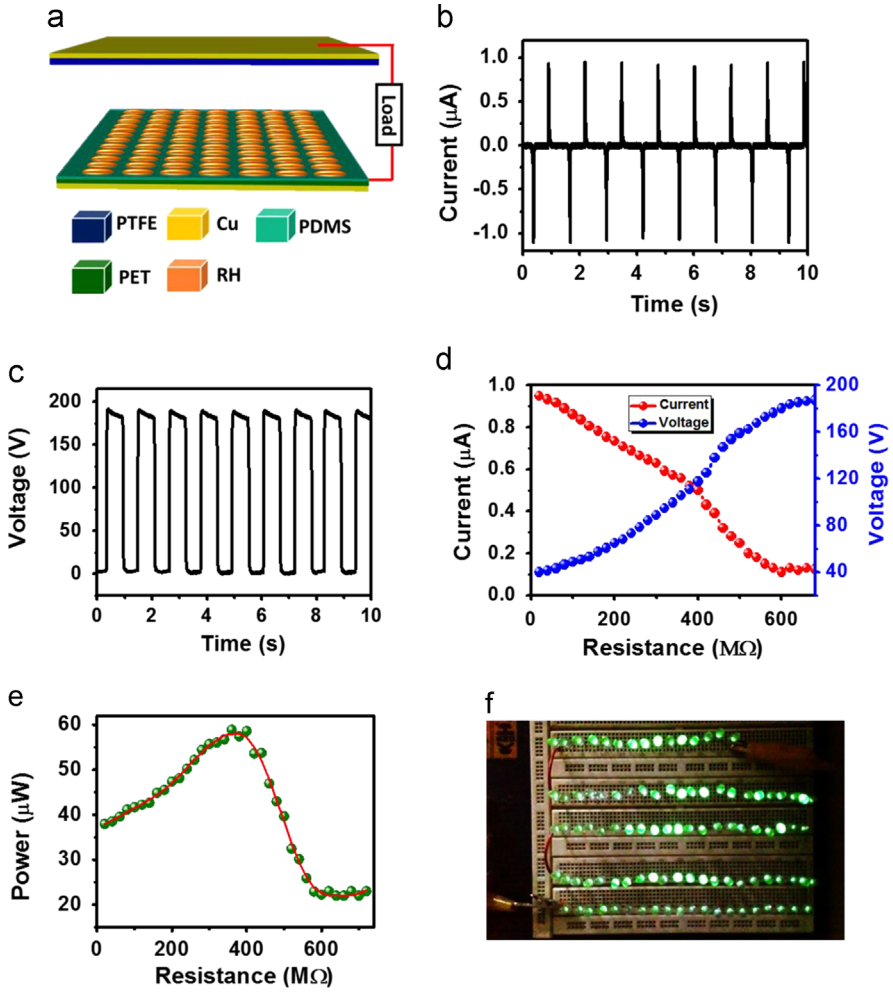


Figure 5 (a) The schematic diagram of the assembled rice-husk arrays TENG (12×7, 84 grains) (b) Short-circuit current of the ARH-TENG. (c) Open-circuit voltage of the ARH TENG. (d) Dependence of the output current and voltage for the single ARH-TENG on the external loading resistance. (e) Output power as a function of variety resistance. (f) The photograph for driving over than 50's LED bulbs by the single ARH-TENG.

further reveals that the output current and voltage of the RH_{SiO2} TENG under a variety of loading resistance are similar with the behavior of Figures 4g and 5d. The largest output instantaneous power of an individual RH_{SiO2}-TENG is 2.13 mW under a loading resistance of $2\times10^8\Omega$, as shown in Figure 6e. The high output power can drive more than 150's LED bulbs (see Figure 6f). The area power density of the individual RH_{SiO2}-TENG is 0.84 W m^{-2} with a peak short-circuit current density of 5.7 mA m^{-2}. Such high current density is almost among the highest reported value based on dielectric-to-dielectric mode TENGs [30,31]. The reason for high short-circuit current density is attributed to that a wide range of pore size is largely distributed throughout an entire RH_{SiO2} film, leading to the rapidly charging and discharging process were generated on the surface of the RH_{SiO2} film.

On the basis of our experimental results, we therefore postulated that the RH_{SiO2} fragments exhibits rich Si-OH bonds (see Figure 7a). To prove that, the FTIR measurement was employed to investigate the stretching bonds of the RH_{SiO2} and the commercial SiO_2 films. The FTIR spectra in Figure 7b evidences that the RH_{SiO2} film presents much strong intensity of the Si-O, OH, Si-OH, and Si-O-Si stretching vibrations when compare with the commercial SiO_2 nanoparticles. The Si-O-Si stretching bonds is one possibility to form the a-SiO:H bonds in a random fashion when H is bonded to an O forming Si-OH [32]. This can further explain how the strongly stretching bonds of the H-O-H and OH have been observed in the RH_{SiO2} film and did not present in the commercial SiO_2 nanoparticles, as shown the FTIR spectra in Figure 7b. Thus, the nanoporous RH_{SiO2} film offers highly dense Si-O-Si and OH bonds (see the schematic diagram in Figure 7c), which is expected to exhibit strongly tendency to repel the electron from their surface because the H atoms have an extremely low electron affinity in contrast to PTFE film. The RH_{SiO2} and PTFE films have strong tendency in donating and receiving electron, respectively. As a consequence, upon the contact of the porous RH_{SiO2} and untreated PTFE films, the charge redistribution between two friction surfaces, the electrons were donated from the porous RH_{SiO2} to the PTFE surfaces because the large difference of electron affinity was established between surfaces (see Figure 7d). The negative and positive charges piled up on the PTFE and RH_{SiO2} surfaces, respectively,

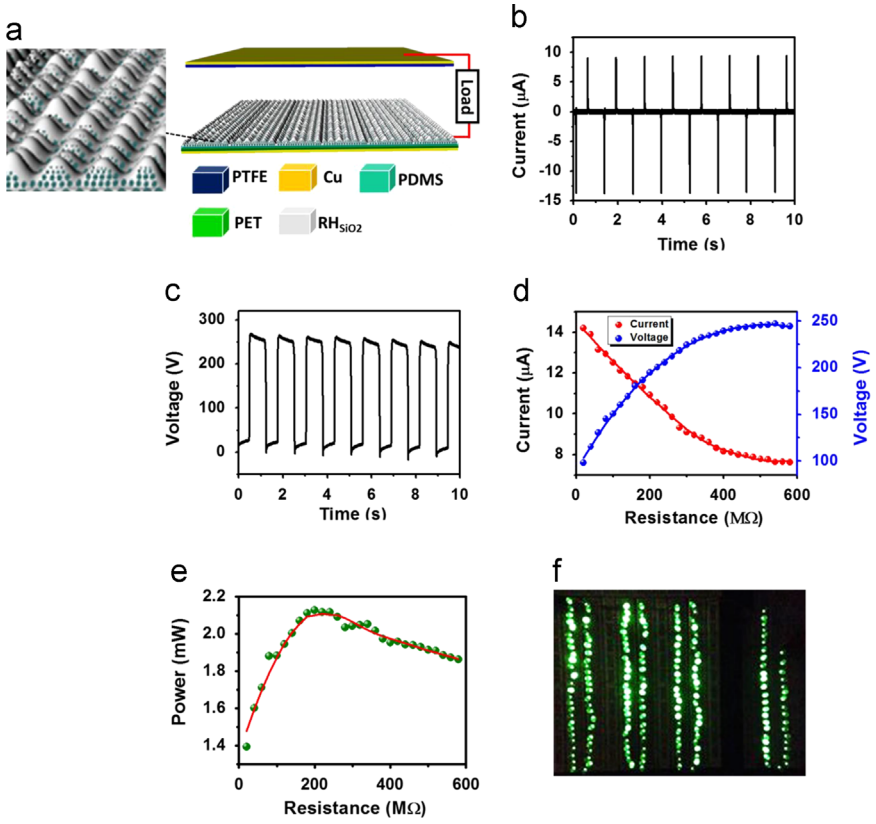


Figure 6 (a) The schematic diagram of the highly porous RH_{SiO2}-TENG. (b) Short-circuit current of the RH_{SiO2} TENG. (c) Open-circuit voltage of the RH_{SiO2} TENG. (d) Dependence of the output current and output voltage of the single RH_{SiO2}-TENG on the external loading resistance. (e) Output power as a function of variety resistance. (f) The photograph for lighting 150's LED bulbs by the single RH_{SiO2}-TENG.

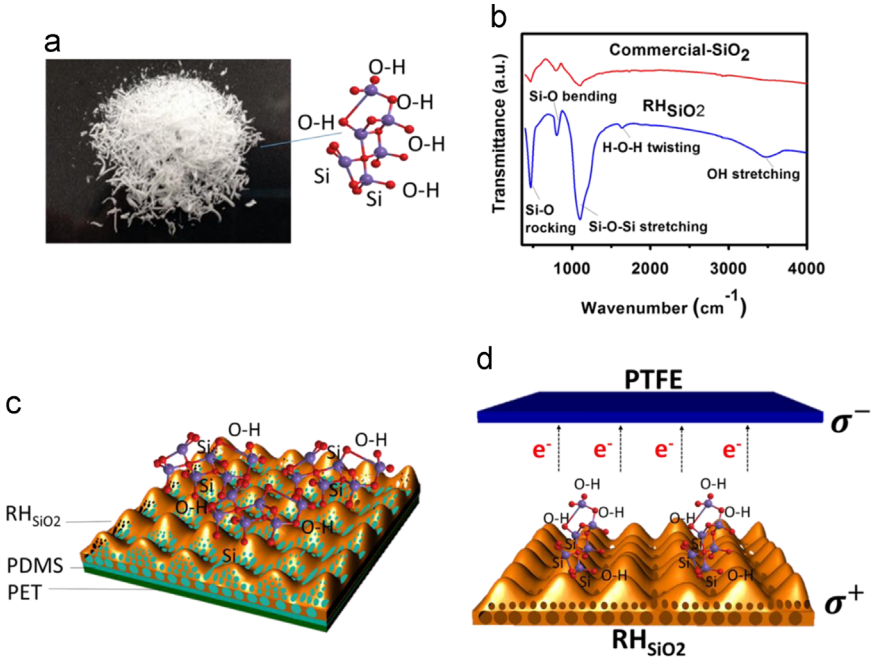


Figure 7 (a) RH_{SiO2} fragments and their crystallographic structure. (b) FTIR spectra of the commercial SiO_2 nanoparticles and RH_{SiO2} fragments. (c) Schematic diagram of the RH_{SiO2} film on the PDMS/PET substrate. (d) Schematic of the charge redistribution between the untreated PTFE and RH_{SiO2} films.

leading to a significantly electrical energy stored between the two surfaces. When the PTFE film was released from the RH_{SiO2}'s surface, the stored electrical energy was generated in the form of an electrical current/voltage through the external circuit. The detail working mechanism is explained in the Supplementary Information of Figure S2. We therefore concluded that the RH_{SiO2} film serves as a very low electron affinity to strongly repel the electrons from its surface and inject into the PTFE film of group-fluoro (F) bonds. The positively charged surfaces in the nanoporous RH_{SiO2} film is dramatically increased compared to the commercial SiO_2 nanoparticles (see Figure 3 above).

To understand the theoretical and transferred charges based on the dielectric-to-dielectric contact mode of RH_{SiO2} TENG, the voltage between two electrodes is given by Eq. (1) [33].

$$V = E_1 t_1 + E_2 t_2 + E_{air} d \tag{1}$$

Where the electric-filed strength at each dielectric layer is given by E_1, E_2, and E_{air}, as expressed by Eqs.. (2), (3), and (4), respectively. t_1 and t_2 are the thickness of the RH_{SiO2} and the PTFE films, respectively. d is the distance between two induced charge surfaces.

$$E_1 = -\frac{Q}{A\varepsilon_{r1}\varepsilon_0} \tag{2}$$

$$E_2 = -\frac{Q}{A\varepsilon_{r2}\varepsilon_0} \tag{3}$$

$$E_{air} = \frac{-Q + A\sigma}{A\varepsilon_0} \tag{4}$$

where A denotes the area of electrode, ε_{r1} and ε_{r2} are relative permittivity of two dielectric layers. ε_0 is vacuum permittivity. σ is the theoretically charge density (see Eq. (6) below). Q is the transferred charges between electrodes. On the basis of (1)-(4), we obtained the Eq. (5) [34].

$$V = \frac{\sigma d}{\varepsilon_0} - \left(\frac{t_1\varepsilon_{r2} + t_2\varepsilon_{r1}\sigma}{\varepsilon_{r1}\varepsilon_{r2}} + d\right)\frac{Q}{A\varepsilon_0} \tag{5}$$

To determine the theoretically charged density (σ) on the electrificated surfaces of the RH_{SiO2} TENG, one can using the following Eq. (6) for calculation. Where V_{oc} is the open-circuit voltage.

$$\sigma = \frac{V_{oc}\varepsilon_0}{d} \tag{6}$$

Thus, the transferred charges (Q) can be obtained by Eq. (7) when two triboelectrically charged surfaces are completely released.

$$Q = \frac{\sigma d\varepsilon_{r1}\varepsilon_{r2}}{t1\varepsilon_{r2} + t2\varepsilon_{r1} + d\varepsilon_{r1}\varepsilon_{r2}} \tag{7}$$

The theoretically charged density (σ) of the RH_{SiO2} reaches 221 $\mu C\,m^{-2}$ if the thickness of the RH_{SiO2} and PTFE films can be reduced to 2 μm. That is, the transferred charge density (Q) of the RH_{SiO2} TENG is estimated ~183 $\mu C\,m^{-2}$. Thus, the ratio of theoretically and transferred charge density (σ/Q) reaches 83%. The debris rice husks are renewable waste and eco-friendly material, which utilizes the most abundantly available agricultural waste. The RH_{SiO2} film can be integrated in various substrates, makes it a valuable material for a large-scale TENG and providing a promising solution for developing a high performance of the dielectric-to-dielectric contact mode TENG.

Conclusion

In summary, we are the first to discover the high output-current TENG based on the dielectric-to-dielectric contact mode using rice husks as a source material. The debris rice husk can directly phase transited into the highly nanoporous SiO_2 structure through the thermally annealing with an additional acid hydrolysis processes. The uniquely porous configuration of RH_{SiO2} possesses highly dense Si-OH and OH stretching bonds, which exhibits very low electron affinity with positively charged surface to repel the electrons from its surface. The area power density of the RH_{SiO2} TENG reaches 0.84 $W\,m^{-2}$ with a peak short-circuit current density of 5.7 $mA\,m^{-2}$, respectively. To the best of our knowledge, the high short-circit current density is almost among the highest reported value based on dielectric-to-dielectric contact mode triboelectric nanogenerators. An individual RH_{SiO2} TENG (area ~5 cm^2) can be used as a direct power source for driving 150 of serially connected LED bulbs. The RH_{SiO2} TENG exhibit exceptionally chemical and thermal stability for superior corrosion resistance, which is opening up new perspective researches in terms of excellent robustness and cost efficiency for green-energy applications.

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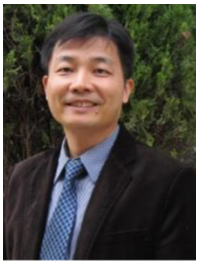
Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.nanoen.2015.11.014.

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